ExamLens

A Private Local VLM Agent for Automatic Exam-Lecture Generation on AMD Radeon GPUs (Track 2)

Problem. Teachers spend hours retyping exam questions and building review slides; exam papers are too sensitive for public cloud APIs.

Solution. A fully-local vision-language-model agent on one AMD Radeon GPU that reads scanned pages, transcribes every question, and renders a complete lecture PowerPoint automatically.

How It Works. 1) VLM Solver (vLLM/ROCm) reads 8 scanned pages and solves 23 questions. 2) VLM Transcriber copies each question verbatim with a page-whitelist anti-hallucination filter. 3) Validator repairs JSON and re-runs missing items until 23/23 coverage. 4) python-pptx renders a 40+ slide deck with original text, answers, explanations, common mistakes and a scanned appendix.

Results. 23/23 questions transcribed; 40+ slides rendered end-to-end; about 2000 tok/s prefill and 30 tok/s decode; zero data egress.

Why AMD Radeon. vLLM ROCm backend, PagedAttention, BF16, graph capture and torch.compile deliver smooth local inference on a single Radeon GPU.